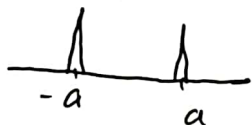


$$(1) \quad E = -\frac{e^2}{2an^2} \quad 2n^2$$

(2) 由 L-S 耦合 自相对论修正

超精细 电子. 质子 偶极矩 磁矩相同

$$(3) \quad E_n = -\frac{Ze^2}{2an^2} = -\frac{(1.2)^2 (\frac{e}{3})^2}{2 \times 0.07 a_0 n^2} \approx -2.3 \frac{e^2}{2a_0 n^2}$$



$$(1) \quad \psi(a_+) = \psi(a_-)$$

$$\psi(-a_+) = \psi(-a_-)$$

$$\psi'(a_+) - \psi'(a_-) = \frac{2mA}{\hbar^2} \psi(a)$$

$$\psi'(-a_+) - \psi'(-a_-) = \frac{2mA}{\hbar^2} \psi(a)$$

(2) $k, \delta B$

$$Be^{-k_1 a} = \cosh k_1 a$$

$$-k_1 Be^{-k_1 a} - k_1 \sinh k_1 a = \frac{2mA}{\hbar^2} Be^{-k_1 a}$$

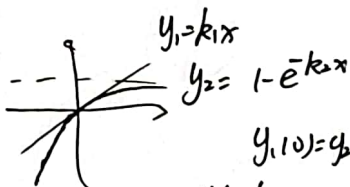
$$-k_1 (\sinh k_1 a + \cosh k_1 a) = \frac{2mA}{\hbar^2} Be^{-k_1 a}$$

$$-k_1 e^{k_1 a} = \frac{2mA}{\hbar^2} Be^{-k_1 a}$$

$$B = \frac{-k_1 \hbar^2}{2mA} e^{2k_1 a}$$

(3) ?

(4)



$$y_1(1) = y_2(1)$$

$x=0$ 时有一解

$$y_1' = k_1$$

$$y_2' = k_2 e^{-k_2 x}$$

$k_1 \neq k_2$ 另有解



三.

$$S_z = \hbar \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix} \quad S_x = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$(1) \quad H = A S_z + B S_x^2$$

$$= A \hbar \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix} + B \frac{\hbar^2}{2} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} A\hbar + \frac{1}{2}B\hbar^2 & 0 & \frac{B}{2}\hbar^2 \\ 0 & B\hbar^2 & 0 \\ \frac{B}{2}\hbar^2 & 0 & -A\hbar + \frac{B}{2}\hbar^2 \end{pmatrix}$$

求特征值

$$\begin{vmatrix} a+b-\lambda & 0 & b \\ 0 & 2b-\lambda & 0 \\ b & 0 & -a+b-\lambda \end{vmatrix} = 0$$

$$(b-\lambda+a)(b-\lambda-a)(2b-\lambda)$$

$$-b^2(2b-\lambda) = 0$$

$$\lambda_1 = 2b \quad \lambda_{2,3} = b \pm \sqrt{b^2 + a^2} \quad \lambda_3 = b - \sqrt{b^2 + a^2}$$

$$E_1 = B\hbar^2$$

$$E_2 = \frac{B}{2}\hbar^2 + \sqrt{(\frac{B}{2}\hbar^2)^2 + (A\hbar)^2}$$

$$E_3 = \frac{B}{2}\hbar^2 - \sqrt{(\frac{B}{2}\hbar^2)^2 + (A\hbar)^2}$$

(2)

~~全~~ ~~错~~

$$E_1 = B\hbar^2 = 2b \quad \psi_1 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

$$E_2 = b + \sqrt{b^2 + a^2} \quad \psi_2 = \frac{1}{\sqrt{2}\sqrt{a^2 + b^2 + a}\sqrt{a+b}} \begin{pmatrix} a + \sqrt{a^2 + b^2} \\ 0 \\ b \end{pmatrix}$$

$$E_3 = b - \sqrt{b^2 + a^2} \quad \psi_3 = \frac{1}{\sqrt{2}\sqrt{a^2 + b^2 - a}\sqrt{a-b}} \begin{pmatrix} a - \sqrt{a^2 + b^2} \\ 0 \\ b \end{pmatrix}$$

$$\psi(0) = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = c_1 \psi_1 + c_2 \psi_2 + c_3 \psi_3$$

$$c_1 = \langle \psi_1 | \psi_0 \rangle =$$

$$c_2 = \langle \psi_2 | \psi_0 \rangle = \frac{a + \sqrt{a^2 + b^2}}{\sqrt{2}\sqrt{a^2 + b^2 + a}\sqrt{a+b}}$$

$$c_3 = \langle \psi_3 | \psi_0 \rangle = \frac{a - \sqrt{a^2 + b^2}}{\sqrt{2}\sqrt{a^2 + b^2 - a}\sqrt{a-b}}$$

$$S_z \text{ 为 } -\hbar \quad \psi' = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$P = |\langle \psi' | \psi(t) \rangle|^2$$

$$= \left| \frac{c_2 e^{-iE_2 t/\hbar}}{\sqrt{2}\sqrt{a^2 + b^2 + a}\sqrt{a+b}} \begin{pmatrix} a + \sqrt{a^2 + b^2} \\ 0 \\ b \end{pmatrix} + \frac{c_3 e^{-iE_3 t/\hbar}}{\sqrt{2}\sqrt{a^2 + b^2 - a}\sqrt{a-b}} \begin{pmatrix} a - \sqrt{a^2 + b^2} \\ 0 \\ b \end{pmatrix} \right|^2$$

$$P = \frac{a - \sqrt{a^2 + b^2}}{b} + \frac{b^2}{2(a^2 + b^2)} \cos$$



④

(1)

$$\int_{-\infty}^{\infty} |\psi|^2 dx = 1$$

$$1 = \int_{-\infty}^{\infty} A^2 e^{-\alpha^2 x^2} dx$$

$$= 2A^2 \int_0^{\infty} e^{-\alpha^2 x^2} dx$$

$$= 2A^2 \cdot \sqrt{\pi} \cdot \frac{0!}{0!} \left(\frac{1}{2\alpha}\right)^{0+1}$$

$$= 2A^2 \frac{\sqrt{\pi}}{2\alpha}$$

$$1 = A^2 \frac{\sqrt{\pi}}{\alpha}$$

$$A = \sqrt{\frac{\alpha}{\pi}}$$

$$P = |\psi|^2 = \frac{\alpha}{\pi} e^{-\alpha^2 x^2}$$

(2)

$$\langle x \rangle = \int_{-\infty}^{\infty} \psi^* x \psi dx = 0$$

$$\langle x^2 \rangle = \int_{-\infty}^{\infty} A^2 x^2 e^{-\alpha^2 x^2} dx$$

$$= 2A^2 \cdot \sqrt{\pi} \frac{2!}{1!} \left(\frac{1}{2\alpha}\right)^3$$

$$= 2 \cdot \frac{\alpha}{\pi} \sqrt{\pi} \cdot 2 \cdot \frac{1}{2\alpha^3}$$

$$= \frac{1}{2\alpha^2}$$

$$\langle p \rangle = \int_{-\infty}^{\infty} \psi^* \frac{\hbar}{i} \frac{\partial}{\partial x} \psi dx$$

$$= \int_{-\infty}^{\infty} i\hbar A^2 \alpha^2 x e^{-\alpha^2 x^2} dx = 0$$

$$\langle p^2 \rangle = \int_{-\infty}^{\infty} \psi^* (-\hbar^2) \frac{\partial^2}{\partial x^2} \psi dx$$

$$= -2A^2 \hbar^2 \alpha^2 \int_0^{\infty} (1 - \alpha^2 x^2) e^{-\alpha^2 x^2} dx$$

$$= \frac{\alpha^2}{2} \hbar^2$$

(3)

$$E_n = (n + \frac{1}{2}) \hbar \omega$$

$$\psi_n = \frac{1}{\sqrt{n!}} (a_+^\dagger)^n \psi_0$$

其中 $a_+ = \frac{1}{\sqrt{2\hbar m \omega}} (\hat{x} - i\hat{p} + m\omega x)$

$$\psi_0 = \frac{\alpha}{\sqrt{\pi}} e^{-\frac{1}{2}\alpha^2 x^2}$$

(4)

$$\langle 0 | x^2 | 0 \rangle = \langle x^2 \rangle - \langle x \rangle^2$$

$$= \frac{1}{2\alpha^2}$$

$$\langle 0 | p^2 | 0 \rangle = \langle p^2 \rangle - \langle p \rangle^2$$

$$= \frac{\alpha^2}{2} \hbar^2$$

$$\langle 0 | x^2 | 0 \rangle \langle 0 | p^2 | 0 \rangle = \frac{\hbar^2}{4}$$



五.

$$a_{\pm} = \frac{1}{\sqrt{2\hbar m \omega}} (\mp i p + m \omega x)$$

六.

$$11) \quad E_n = E_n^{(0)} + H_{nn}' + \sum_{m \neq n} \frac{|H_{mn}'|^2}{E_n - E_m}$$

$$E_1 = 1 \text{ 级 } E_1^{(1)} = H_{11}' = \langle \psi_1 | H' | \psi_1 \rangle$$

$$2 \text{ 级 } E_1^{(2)} = \frac{|\langle \psi_1 | H' | \psi_2 \rangle|^2}{E_1 - E_2}$$

$$E = E_1^{(0)} + E_1^{(1)} + E_1^{(2)}$$

$$\psi_n = \psi_n^{(0)} + \sum_{m \neq n} \frac{H_{mn}'}{E_n - E_m} \psi_m$$

$$\psi_1 = |\psi_1\rangle + \frac{\langle \psi_2 | H' | \psi_1 \rangle}{E_1 - E_2} |\psi_2\rangle$$

E_2

$$1 \text{ 级 } E_2^{(1)} = H_{22}' = \langle \psi_2 | H' | \psi_2 \rangle$$

$$2 \text{ 级 } E_2^{(2)} = \frac{|\langle \psi_1 | H' | \psi_2 \rangle|^2}{E_2 - E_1}$$

$$E_2 = E_2^{(0)} + E_2^{(1)} + E_2^{(2)}$$

$$\psi_2 = |\psi_2\rangle + \frac{\langle \psi_1 | H' | \psi_2 \rangle}{E_2 - E_1} |\psi_1\rangle$$

$$(2) \quad H' = \begin{pmatrix} \alpha & 0 \\ 0 & \alpha \end{pmatrix} \quad E_1 = E_2 = E \quad \text{二重简并}$$

$$\text{令 } \psi_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \psi_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$H' \text{ 由已知显然 } H_{11} = H_{22} = \alpha \quad H_{12} = H_{21}^* = 0$$

$$\text{久期} \quad \begin{pmatrix} H_{11} - E^{(1)} & 0 \\ 0 & H_{22} - E^{(1)} \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = 0$$

$$\text{得} \quad E^{(1)} = \alpha$$

$$E = E_1 + \alpha \quad (E_1 = E_2)$$

$$|\psi_1\rangle \quad |\psi_2\rangle \quad \text{未变}$$

